NAFLD Liver Fibrosis Risk Estimation

Comprehensive Technical Documentation: Frontend Architecture, Engineering & Operational Workflow

Project: Major Project / Capstone Research	**Target Domain:** Non-Invasive Hepatology ML
Backend Service: Python FastAPI (Uvicorn)	**Frontend Framework:** React 19 + TypeScript + Vite
ML Model Bundle: Tuned Random Forest (n=300, leaf=5)	**Decision Threshold:** 0.4158 (>=80% Dev Sensitivity)
Clinical Predictors: 33 Approved Routine Lab Parameters	**Cohort Size:** N = 605 Biopsy-Proven NAFLD Patients
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1. Executive Summary & Clinical Problem Formulation

Non-Alcoholic Fatty Liver Disease (NAFLD) represents the most common chronic liver pathology worldwide, affecting over 25% of global adults. While simple steatosis (fat accumulation) follows a relatively benign course, progression to **significant fibrosis (histologic stage F ≥ 2)** substantially escalates the risks of cirrhosis, hepatic failure, and hepatocellular carcinoma.

The Clinical Diagnostic Bottleneck: Percutaneous liver biopsy remains the definitive diagnostic gold standard for histologic staging. However, biopsy is invasive, carries a 1-3% complication risk (hemorrhage, perforation, biliary peritonitis), suffers from significant inter-observer variability, and is unsuitable for wide population screening.

Inadequacy of Existing Non-Invasive Scores: The clinical benchmark FIB-4 (Fibrosis-4 index) relies on only four features: Age, AST, ALT, and Platelets. In our 605-patient biopsy-verified cohort, FIB-4 achieves an area under the ROC curve of **0.703**, but at the conventional 1.30 rule-out cutoff, its sensitivity is only **50.2%**—meaning approximately half of patients with significant fibrosis are missed.

System Objective: Build an end-to-end, zero-leakage research platform that leverages 33 routine laboratory and clinical parameters. The frontend is engineered with an editorial medical journal aesthetic (Nature/NEJM style) providing interactive clinical estimation, subgroup fairness visualization, benchmark comparisons, and transparent methodology.

2. Frontend Technology Stack & Design Architecture

The frontend client is built as a single-page application (SPA) optimized for zero-latency form interaction, medical data integrity, and dynamic data visualization.

Category	Technology / Library	Purpose in Dashboard Implementation
Core UI Engine	**React 19** (TypeScript)	Component lifecycle, modular clinical panels, reactive state rendering.

Build & Bundler	**Vite** (Rollup)	Instant sub-second Hot Module Reload (HMR) and optimized minified SPA distribution.
Styling System	**Tailwind CSS 3.4**	Custom design tokens (sage, canvas, deep slate, risk alerts) ensuring medical publication aesthetics.
Data Visualizations	**Recharts**	Interactive SVG charting: ROC curve comparison, subgroup fairness bars, and feature importances.
Form Management	**React Hook Form**	Uncontrolled form inputs preventing whole-page re-renders across all 33 medical input fields.
Schema Validation	**Zod** & Resolvers	Enforces cohort-derived physiological boundaries (e.g. valid AST, ALT, BMI, Platelet bounds).
Iconography & Fonts	**Lucide React** + Google Fonts	Source Serif 4 for academic headings; Inter for UI; JetBrains Mono for metrics and thresholds.

3. Detailed Component Architecture & Screen Breakdown

The interface is organized into a modular component hierarchy located in frontend/src/. Each view serves a targeted clinical and research purpose:

3.1 Persistent Journal Masthead (Header.tsx)

- **Live Health Ping:** On component mount, the client polls GET /health. When online, it renders a green indicator (*FastAPI Online*). If the backend is disconnected, it activates the fallback client engine (*Bundled Engine*).
  - **Cohort & Model Badges:** Displays cohort volume (*N=605*), hyper-parameters (*Tuned RF: n=300, leaf=5*), and the research threshold (*0.4158*).
  - **Ethical Governance:** Prominently states: *'Academic Research Prototype • Not approved as a medical device or diagnostic instrument'* to prevent diagnostic misuse.

3.2 Clean Tabbed Navigation (Navigation.tsx)

Provides seamless single-page switching between four primary analytical modules without page reloading: **Overview**, **Risk Estimator (33 Inputs)**, **Model Performance (ROC & Subgroups)**, and **Data & Method**.

3.3 Overview & Executive Summary (Overview.tsx)

- **Key Metric Cards:** Displays Tuned RF ROC-AUC (**0.738**), Sensitivity (**80.5%**), Specificity (**65.0%**), and Decision Threshold (**0.4158**).
  - **Benchmark Contrast:** Compares performance against clinical FIB-4 (AUC 0.703, Sensitivity 50.2%).
  - **Cohort Histologic Distribution:** Details the baseline biopsy composition of the 605 patients: 386 mild cases (F0-1, 63.8%), 111 significant fibrosis cases (F2, 18.3%), and 108 advanced fibrosis cases (F3-4, 17.9%).

3.4 Interactive Risk Estimator (RiskEstimator.tsx)

This is the central operational tool of the application, designed for rapid clinical data entry and transparent probabilistic interpretation:

Panel Group	Clinical Variables Included	Physiological Rationale
1. Demographics & History	Age, Biological Sex, Smoking History	Fibrosis prevalence increases markedly with age (>50) and active smoking.
2. Body Measures & Vitals	Body Mass Index (BMI), Waist Circumference, Systolic BP, Diastolic BP	Visceral adiposity and hypertension directly correlate with insulin resistance.
3. Liver Biochemistry	ALT, AST, ALP, GGT, Total Bilirubin, Direct Bilirubin, Total Protein, Albumin	Enzymatic necrosis markers (AST/ALT ratio) and synthetic liver capacity (Albumin).
4. Hematology & Coagulation	Platelets (Trombosit), WBC, RBC, Hemoglobin, Hematocrit, MCV, MCH, MCHC, RDW	Thrombocytopenia (low platelets) reflects portal hypertension and splenic sequestration.
5. Metabolic & Renal Panel	Fasting Glucose, HbA1c, Triglycerides, Total Cholesterol, HDL, LDL, BUN, Creatinine	Type-2 Diabetes is the single strongest univariate comorbidity for progressive fibrosis.

3.5 Key Visual Sub-Components in the Estimator

- **Risk Gauge (RiskGauge.tsx):** A calibrated radial arc gauge visualizing continuous probability from 0% to 100%. Color-coded into three clinical tiers: Green (0–25% Low Risk), Amber (25–41.6% Indeterminate), and Crimson (≥41.6% Elevated Risk). A distinct marker flags the 0.4158 operating threshold.
- **Parallel FIB-4 Comparator (FIB4Card.tsx):** Automatically computes the patient's FIB-4 score in parallel using formula: (Age × AST) / (Platelets × √ALT). Displays clinical cutoffs: <1.30 (Rule-Out), 1.30–2.67 (Indeterminate), >2.67 (Rule-In).
- **1-Click Clinical Presets:** Features instant loading buttons for *'Typical Low-Risk Patient (F0-F1)'*, *'Typical Elevated-Risk Patient (F2+)'*, and *'Cohort Median Patient'* for immediate live testing.
- **Out-of-Range Clinical Validation:** Evaluates inputs against 605-patient physiological limits and renders amber advisory banners if extreme outliers are detected.

3.6 Model Performance & Transparency (ModelPerformance.tsx)

- **Interactive ROC Curve:** Plots Tuned Random Forest against FIB-4 clinical score, visually showing superior true-positive rates across sensitivity thresholds.
- **Confusion Matrix (ConfusionMatrix.tsx):** Displays out-of-fold development counts: True Negatives (251), False Positives (135), False Negatives (35), and True Positives (144).
- **Subgroup Fairness Analysis:** Assesses sensitivity/specificity across age (<50 vs ≥50), biological sex, and diabetes status, ensuring the algorithm does not exhibit systematic clinical bias.
- **Feature Importance:** Displays top predictors derived from permutation importance: Age, AST, Platelets, Diabetes, BMI, and Fasting Glucose.

3.7 Rigor & Methodology (DataMethod.tsx)

- **Data Leakage Prevention:** Documents total exclusion of biopsy-derived columns (Steatosis, Activity, Fibrosis, NAS), helper labels, and FIB-4 from the 33 inputs.
- **Informative Missingness Control:** Documents why 15 extended lab variables (missing >45%) were excluded rather than imputed to avoid selection bias.

4. End-to-End Execution Lifecycle & Data Flow

The following structured sequence traces the end-to-end data lifecycle when a clinician or researcher estimates patient risk on the frontend:

Step	Stage	Technical Execution Description
1	**User Input / Preset**	User inputs clinical parameters or clicks a preset profile in RiskEstimator.tsx.
2	**Client Validation**	react-hook-form and Zod validate numerical types and boundaries without re-rendering the DOM.
3	**API Serialization**	lib/api.ts packages 33 variables into a clean JSON dictionary payload: { features: { Age: 47, AST: 34, ... } }.
4	**FastAPI Inference**	Backend POST /predict feeds features into the serialized Scikit-learn Pipeline (Median Imputer → Standard Scaler → Random Forest).
5	**Probability Estimation**	Model runs predict_proba, evaluates threshold 0.4158, computes FIB-4 score, and returns formatted JSON response.
6	**Reactive UI Update**	Frontend receives payload, triggers smooth needle animation on RiskGauge, and updates classification badges.

5. Key Frontend Engineering & Performance Optimizations

1. **Zero-Lag Uncontrolled Forms:** Managing 33 simultaneous state hooks with standard React useState would cause 33 re-renders per keystroke. By utilizing react-hook-form with uncontrolled refs, typing is instant with zero lag.
2. **Unified Fullstack Single-Service Deployment:** The production frontend is pre-compiled into frontend/dist/. The FastAPI backend mounts these static files using StaticFiles and provides SPA catch-all routing. This allows the entire fullstack application to run on a **single free Render Web Service** with zero CORS issues.
3. **Dual Engine Resilience:** If the backend server sleeps or is disconnected, the frontend does not crash. It automatically falls back to an internal deterministic scoring engine and alerts the user.
4. **Strict TypeScript Contracts:** The 33 feature keys in the frontend types.ts exactly mirror the feature schema generated by Scikit-learn, preventing field-name mismatch errors during deployment.

6. Examiner & Teacher Viva Q&A; Defense Guide

Q: Why was threshold 0.4158 selected instead of the default 0.50?

A: In clinical triage of progressive fibrosis, a False Negative (missing a patient with liver scarring) is far more hazardous than a False Positive. The default 0.50 threshold yielded only ~64% sensitivity. We calibrated the threshold to 0.4158 on nested out-of-fold cross-validation to guarantee ≥ 80% sensitivity (achieving 80.5%), drastically reducing missed at-risk patients.

Q: Why did you predict fibrosis stage (F ≥ 2) rather than fatty liver presence?

A: The source dataset contains 605 patients with biopsy-proven NAFLD. Every single patient already has steatosis (Grade 1, 2, or 3) with zero healthy Grade 0 controls. Therefore, a fatty liver detector cannot be trained. What does vary clinically is fibrosis scarring, which dictates cirrhosis risk and patient survival.

Q: How did you ensure data leakage was avoided between frontend and model?

A: During exploratory data analysis (EDA), all post-biopsy columns (Kleiner NAS, SAF diagnosis, biopsy grade, AtRiskNASH flags, and FIB-4 formula components) were quarantined. Only 33 pre-biopsy routine blood tests and vitals are accepted by the frontend form.